From Waste to Loop: Circular Bottles

New bottle framework regulation for non-food household containers in the scope of Norwegian and European markets

5,3 Mt of HDPE-Plastics are Lost Today

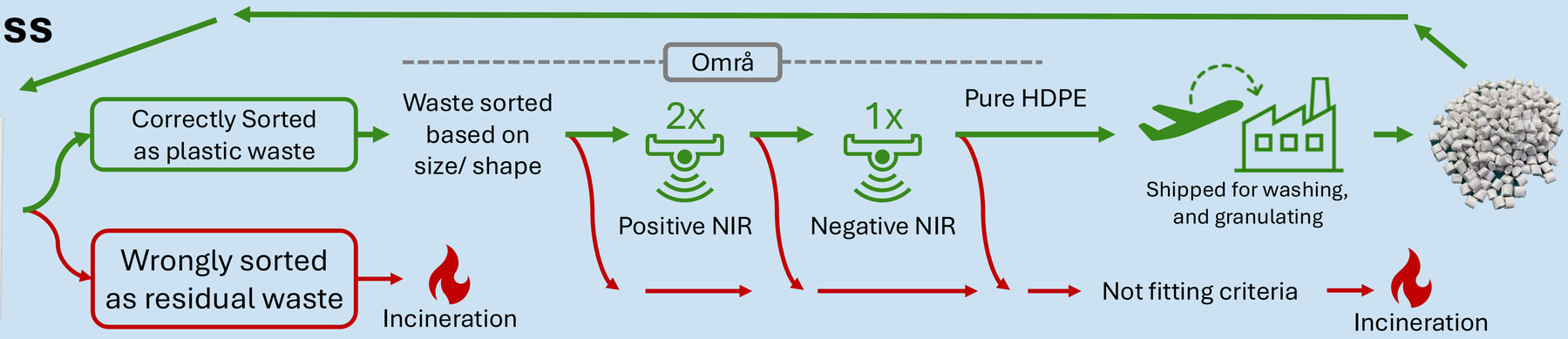
- 6,2 Mt HDPE is yearly consumed in the EU [8]
  - 2,5 Mt are used for bottles [8]
- HDPE is highly recyclable [5]
- Only 10-15% of EU HDPE is recycled today [1]
- HDPE is commonly found as ocean waste [9]

Our Standard of Tomorrow

- Maximize bottle recycling by:
  - Designing for circularity
  - Making it recognizable as recyclable for the consumer
- Enforce regulation

HDPE Recycling Process

Områ, Norway's top plastic sorting facility, processes 90k tons yearly; HDPE sorting is explained here with household waste.



Our Proposed Regulation

Material
- ✓ RecyClass compliant natural or white HDPE [3]

Emptying and Recyclability
- ✓ Recyclable **10+** times [5]
- ✓ Recycled content at least PPWR requirement [2]
- ✓ Design for easy draining
- ✓ Consumer convenience by cleaner recycling process
- ✓ Use standardized recycling-grade additives [3]
- ✓ **NO dark colors:** For good NIR sensor detection: white / grey [3]

Weight-to-Volume Optimization [6]
- ✓ Minimize material → reduce CO_2 emissions during transport
- ✓ Use potential of wall thickness reduction

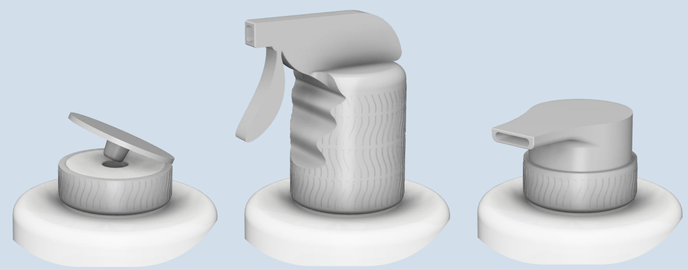
Structural Requirements
- ✓ Rigidity maintained despite thin walls
- ✓ Chemical resistance: no-cross-contamination

Manufacturing [4]
- ✓ Apply Injection Stretch Blow Molding (ISBM)
- ✓ Reduces weight up to 40%

Cap Variety

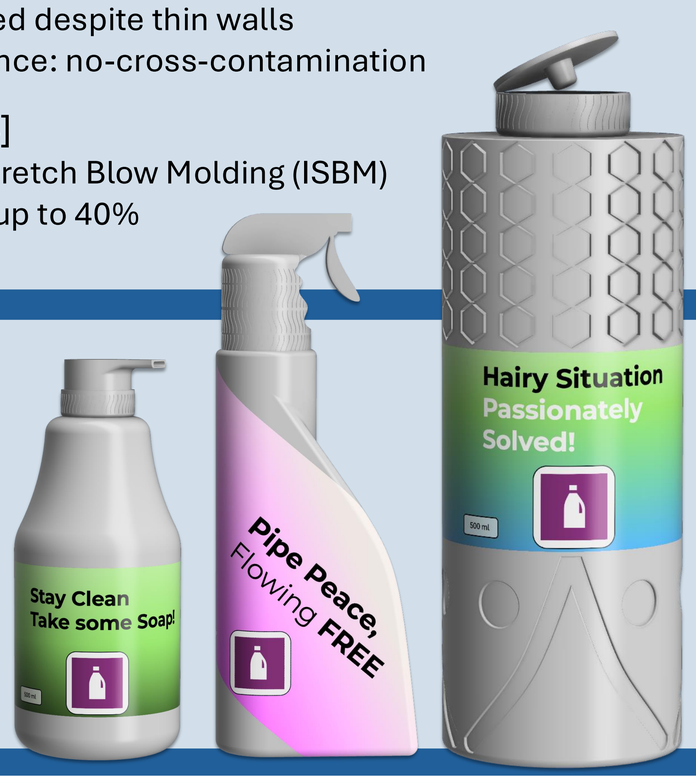
Natural HDPE material same as bottle body

Enable product dependent closures



Design Freedom

Room for brand's signature bottle shapes and labels



Label Design Flexibility

Material
RecyClass compliant **PE (Polyethylene) label film [3]**

Attachment
Durable, non-dissolving in water; easy to dismantle by mechanical or chemical separation

Print
Non-bleeding EuPIA-compliant inks [3]

Recycling Instructions
- Clearly placed on the front side
- Large, high-contrast icons and text for visual impairments
- Explain exactly **how** and **where to recycle**

Compliance with EU Packaging and Packaging Waste Regulation 2025/40 (PPWR) [2]

All packaging must be economically recyclable	Increase the recycled content according to milestones	Packaging sector must achieve climate neutrality
2030	**2031-2049**	**2050**

- **Clear Labelling:** Mandatory recycling instructions on all packaging
- **EN Compliance:** Suppliers must verify adherence to packaging standards
- **Virgin Plastic Reduction:** Prioritize recycled content

Summary

The proposed regulation for Circular Bottles shows that design-for-recycling can coexist with brand identity. By standardizing materials, colors, labels and caps, and optimizing wall thickness and emptying behavior, the bottles are designed for high-quality bottle-to-bottle HDPE recycling within existing European systems.

It is expected that brands may resist perceived limitations on design freedom, even when recyclability improves. Consumer behavior remains critical, as incorrect sorting contributes to loss of material and poor emptying will reduce material quality. Circular bottles aim to tackle limiting factors in current HDPE-bottle recycling, looking at the complete lifetime starting from the producer, the consumer, and ending with the recycling facility.

More Info:

Solveig Aarmo, Sunniva Belsnes, Jonas Hain, Agnes Cecilia Scheel Melin, Kristoffer Skarsvåg Midling and Finja Kathleen Ohm – *TEP4854: Experts in Teamwork – Plastic Free Oceans 2026*